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Secondary battery

Abstract

An embodiment of the present invention relates to a secondary battery, and the problem to be solved by the present invention is to provide a secondary battery in which output is improved by minimizing component resistance or internal resistance, and in which material costs may be reduced. To this end, disclosed is a secondary battery comprising: a cylindrical can having an opening; an electrode assembly accommodated in the opening of the cylindrical can and wound in a cylindrical form; a current collector plate electrically connected to the electrode assembly; and a safety vent electrically connected to the current collector plate and sealing the opening of the cylindrical can.

Inventors:	Kim; Mun Sung (Yongin-si, KR)
Applicant:	Samsung SDI Co., Ltd. (Yongin-si, KR)
Family ID:	73458322
Assignee:	Samsung SDI Co., Ltd. (Yongin-si, KR)
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Primary Examiner: Douyette; Kenneth J

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is a U.S. National Phase Patent Application of International Application Number PCT/KR2020/006547, filed on May 19, 2020, which claims priority of Korean Patent Application Number 10-2019-0060097, filed on May 22, 2019, the entire content of each of which is incorporated herein by reference.

TECHNICAL FIELD

(2) An embodiment of the present invention relates to a secondary battery.

BACKGROUND ART

(3) Lithium ion secondary batteries are widely used, for example, as power sources for hybrid vehicles or electric vehicles as well as for portable electronic devices, due to certain characteristics including having a high operating voltage and a high energy density per unit weight.

(4) The secondary batteries may be classified as a cylindrical type, a prismatic type, or a pouch type. Among these secondary batteries, the cylindrical secondary battery generally includes a cylindrical electrode assembly, a cylindrical can to which the electrode assembly is coupled, an electrolyte injected into the can so as to allow lithium ions to move, a cap assembly coupled to one side of the can to prevent leakage of the electrolyte and to prevent separation of the electrode assembly, and the like.

(5) The above information disclosed in this section of BACKGROUND ART is only for enhancement of understanding of the background of the present invention, and therefore it may contain information that does not form the related art.

DISCLOSURE OF THE INVENTION

Technical Problem

(6) An embodiment of the present invention provides a secondary battery in which output is improved by minimizing component resistance or internal resistance, and in which material costs may be reduced.

Technical Solution

(7) A secondary battery according to an embodiment of the present invention includes: a cylindrical can having an opening; an electrode assembly accommodated in the opening of the cylindrical can and wound in a cylindrical form; a current collector plate electrically connected to the electrode assembly; and a safety vent electrically connected to the current collector plate and sealing the opening of the cylindrical can.

(8) An electrode tab, which has a flat top surface and a flat bottom surface on the opposite side from the top surface, may be further interposed between the safety vent and the current collector plate.

(9) A central region of the top surface of the electrode tab may be welded to the safety vent, and an edge region of the bottom surface of the electrode tab may be welded to the current collector plate.

(10) The current collector plate may further include a central through-hole formed in a central region.

(11) The current collector plate may further include a peripheral through-hole formed on the outside of a central region.

(12) The secondary battery according to an embodiment of the present invention may further include a cap-up, and the cap-up may be electrically connected to an upper part of the safety vent.

(13) The safety vent may further include a bent part which is bent upward along an edge and coupled to an edge of the cap-up.

(14) The cap-up may include: an edge flat part electrically connected to the safety vent; a bent part bent upward from the edge flat part to the inside; and a central flat part extending horizontally from the bent part and spaced apart from the safety vent.

(15) The bent part further may include a peripheral through-hole.

(16) The central flat part may further include a central through-hole.

Advantageous Effects

(17) The embodiment of the present invention may provide the secondary battery in which the output is improved by minimizing component resistance or internal resistance, and in which the material costs may be reduced. In the electronic elements, the safety vent and the current collector plate are electrically connected to each other through the electrode tab having a flat plate shape, and accordingly, the component resistance or the internal resistance of the secondary battery may be minimized.

(18) Furthermore, as the internal resistance is minimized, the output of the secondary battery may be enhanced, and the heat generation phenomenon of the secondary battery may be reduced as well. Also, since a current interrupting device, an insulating ring, or the like according to the related art is not required, the material costs of the secondary battery may also be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A and 1B are a perspective view and a cross-sectional view illustrating a secondary battery according to an embodiment of the present invention.

(2) FIG. 2A is an enlarged cross-sectional view illustrating a cap assembly in a secondary battery according to an embodiment of the present invention, and FIG. 2B is a perspective view illustrating relationship between a current collector plate and an electrode tab.

(3) FIGS. 3A to 3F are views for explaining input/output of current, and various contact surfaces and welding surfaces in a secondary battery according to an embodiment of the present invention.

MODES FOR CARRYING OUT THE INVENTION

(4) Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings.

(5) The embodiments of the present invention are provided to more fully describe the present invention to those skilled in the art. The following embodiments may be modified in many different forms, and the scope of the present invention is not limited to the following embodiments. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the concept of the present invention to those skilled in the art.

(6) Also, the thickness and size of each layer are exaggerated in the drawings for convenience of explanation and clarity, and like reference numerals in the drawings refer to like elements throughout. As used in this specification, the term “and/or” includes any and all combinations of one or more of the associated listed items. Also, in this specification, it will be understood that when an element A is referred to as being “connected to” an element B, the element A can be directly connected to the element B, or an intervening element C may be present between the elements A and B so that the element A can be indirectly connected to the element B.

(7) The terminologies used herein are for the purpose of describing particular embodiments only and are not intended to be limiting of the present invention. As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. Also, it will be further understood that the terms “comprise or include” and/or “comprising or including,” when used in this specification, specify the presence of stated features, numbers, steps, operations, members, elements, and/or groups thereof, but do not preclude the presence or addition of one or more other features, numbers, steps, operations, members, elements, and/or groups thereof.

(8) It will be understood that, although the terms first, second, etc. may be used herein to describe various members, components, regions, layers, and/or portions, these members, components, regions, layers, and/or portions should not be limited by these terms. These terms are only used to distinguish one member, component, region, layer, or portion from another region, layer, or portion. Thus, a first member, component, region, layer, or portion which will be described may also refer to a second member, component, region, layer, or portion, without departing from the teaching of the present invention.

(9) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for easy understanding of one element or feature and another element(s) or feature(s) as illustrated in the drawings. These spatially relative terms are intended for easy comprehension of the present invention according to various process states or usage states of the present invention, and thus, the present invention is not limited thereto. For example, when an element or feature in the drawings is turned over, the element and feature described as “beneath” or “below” are changed into “above” or “upper”. Thus, the term “below” may encompass the term “above” or “below”.

(10) FIGS. 1A and 1B are a perspective view and a cross-sectional view illustrating a secondary battery according to an embodiment of the present invention.

(11) As illustrated in FIGS. 1A and 1B, a secondary battery **100** according to an embodiment of the present invention may include a cylindrical can **110**, an electrode assembly **120** having a cylindrical form, a first current collector plate **130**, a second current collector plate **140**, and a cap assembly **150**.

(12) Also, the present invention may further include an electrode tab **160** having a thin plate shape and electrically connecting the first current collector plate **130** to the cap assembly **150**. Also, the present invention may further include an insulating gasket **170** that insulates the cylindrical can **110** from the cap assembly **150**.

(13) The cylindrical can **110** includes a bottom part **111** having an approximately circular shape and a side part **112** extending a certain length upward from the bottom part **111**. During a manufacturing process of the secondary battery, the upper part of the cylindrical can **110** is open. Thus, during the manufacturing process of the secondary battery, the electrode assembly **120**, the first current collector plate **130**, and the second current collector plate **140** may be integrated into a single structure and inserted into the cylindrical can **110**. Of course, subsequently, an electrolyte

may be additionally injected into the cylindrical can **110**.

(14) The cylindrical can **110** may be made of steel, a steel alloy, nickel plated steel, a nickel plated steel alloy, aluminum, an aluminum alloy, or an equivalent thereof, but the materials are not limited thereto. Furthermore, in order to prevent the cap assembly **150** from being separated to the outside, the cylindrical can **110** may have a beading part **113** recessed inward at the lower part thereof with respect to the cap assembly **150** and a crimping part **114** bent inward at the upper part thereof.

(15) The electrode assembly **120** having a cylindrical shape may be accommodated in the cylindrical can **110**. The electrode assembly **120** includes a first electrode plate **121** coated with a first electrode active material (e.g., a positive electrode active material such as a transition metal oxide (LiCoO_2 , LiNiO_2 , or LiMn_2O_4)), a second electrode plate **122** coated with a second electrode active material (e.g., a negative electrode active material such as graphite, carbon, or silicon), and a separator **123** positioned between the first electrode plate **121** and the second electrode plate **122** and allowing only lithium ions to move therebetween while preventing an electrical short circuit therebetween. The first electrode plate **121**, the second electrode plate **122**, and the separator **123** are stacked and then wound in an approximately cylindrical form. Here, although not limited, for example, the first electrode plate **121** may include an aluminum (Al) foil, and the second electrode plate **122** may include a copper (Cu) or nickel (Ni) foil. Also, the separator **123** may include, but not limited to, a polyethylene separator (PES), a polypropylene separator (PPS), a ceramic coated separator (CCS), a polymer coated separator (PCS), a multi-layer coated separator (MCS), or a multi-functional separator (MFS).

(16) Meanwhile, a first non-coating part **124**, which is not coated with the first electrode active material and protrudes and extends a certain length upward, may be provided in the first electrode plate **121**, and a second non-coating part **125**, which is not coated with the second electrode active material and protrudes and extends a certain length downward, may be provided in the second electrode plate **122**. Here, the directions, in which the first non-coating part **124** and the second non-coating part **125** protrude and extend, are the same as the longitudinal direction of the cylindrical can **110** and/or the electrode assembly **120** having the cylindrical shape. When the direction in which the first non-coating part **124** protrudes and extends is defined as a first direction, the direction in which the second non-coating part **125** protrudes and extends may be defined as a second direction opposite to the first direction.

(17) The first non-coating part **124** and the second non-coating part **125** serve as paths for a charging/discharging current, and the first non-coating part **124** and the second non-coating part **125** directly extend outward (in the cross-section, illustrated as several segments) over the entire width or entire length of the first electrode plate **121** and the second electrode plate **122**, respectively. Thus, deformation (i.e., degradation in the degree of circularity) or cracking of the electrode assembly **120** may be prevented, and there may be a benefit to high output of the battery. In addition, the first non-coating part **124** extends from the first electrode plate **121** and thus may be the same aluminum or aluminum alloy as the first electrode plate **121**. The second non-coating part **125** extends from the second electrode plate **122** and thus may be the same copper, copper alloy, nickel, or nickel alloy as the second electrode plate **122**.

(18) Also, the first non-coating part **124** of the electrode assembly **120** may be electrically connected (welded) to the first current collector plate **130** which will be described later, and the second non-coating part **125** of the electrode assembly **120** may be electrically connected (welded) to the second current collector plate **140** which will be described later. In addition, the first current collector plate **130** may be electrically connected (welded) to the cap assembly **150**, and the second current collector plate **140** may be electrically connected (welded) to the cylindrical can **110**. Thus, the cap assembly **150** may operate as a positive electrode, and the cylindrical can **110** may operate as a negative electrode. Of course, the connection relationships thereof may be reversed. Thus, the cap assembly **150** may operate as a negative electrode, and the cylindrical can **110** may operate as a positive electrode.

(19) The first current collector plate **130** plays a role in electrically connecting the first non-coating part **124** provided in the electrode assembly **120** to the cap assembly **150**. The first current collector plate **130** may have an approximately circular plate shape, and further include a recessed part **131** which compresses the first non-coating part **124** so as to be welded to the first non-coating part **124**. Also, the first current collector plate **130** may further include a central through-hole through which a welding rod passes so that the second current collector plate **140** may be resistance-welded to the bottom part **111** of the cylindrical can **110** and a peripheral through-hole through which an electrolyte may be injected. The first current collector plate **130** may be aluminum or an aluminum alloy.

(20) Meanwhile, the first current collector plate **130** may further include an insulating layer (e.g., polyimide, polypropylene, polyethylene, or metal oxide layer, etc.) which is formed on a side circumferential surface that faces a side wall **112** of the cylindrical can **110**. Substantially, the insulating layer prevents the first current collector plate **130** from being electrically short-circuited to the side wall **112** of the cylindrical can **110**. Here, when the first current collector plate **130** is an aluminum-based material, the insulating layer described above may be an anodizing layer, i.e., an oxide coating or an aluminum oxide layer (Al_2O_3). The thickness of the insulating layer may be, for example, but not limited to, about 10 μm to about 100 μm .

(21) The second current collector plate **140** may electrically connect the second non-coating part **125** provided in the electrode assembly **120** to the cylindrical can **110**. The second current collector plate **140** may have an approximately circular plate shape, and further include a recessed part **141** which compresses the second non-coating part **125** so as to be welded to the second non-coating part **125**. The second current collector plate **140** may be resistance-welded to the bottom part **111** of the cylindrical can **110** by the welding rod. The second current collector plate **140** may be copper, a copper alloy, nickel, or a nickel alloy.

(22) The cap assembly **150** may include a safety vent **151**. Also, the cap assembly **150** may further include a cap-up **152** positioned on the safety vent **151**. In embodiments, a circumferential region of the safety vent **151** is bent to surround the circumference of the cap-up **152**. The safety vent **151** and/or the cap-up **152** may be aluminum or an aluminum alloy.

(23) The electrode tab **160** has an approximately flat plate shape, and may be interposed between the first current collector plate **130** and the safety vent **151**. That is, the electrode tab **160** may electrically connect the first current collector plate **130** to the safety vent **151**.

(24) The insulating gasket **170** surrounds the safety vent **151** and the cap-up **152** while forming an approximately circular ring shape, and plays a role in electrically insulating the safety vent and the cap-up from the side part **111** of the cylindrical can **110**. The insulating gasket **170** has a substantially compressed shape between the beading part **113** and the crimping part **114** which are formed in the side part **111** of the cylindrical can **110**. The insulating gasket **170** may include, for example, but not limited to, a thermal resistance resin. The thermal resistance resin may include, for example, but not limited to, one or two or more selected from the group consisting of polypropylene (PP), polyethylene (PE), polyimide (PI), polybutyleneterephthalate (PBT), polycarbonate (PC), and polystyrene (PS).

(25) Meanwhile, an electrolyte (not illustrated in the drawing) may be injected into the cylindrical can **110**, and the electrolyte plays a role in allowing lithium ions to move, the lithium ions being generated through an electrochemical reaction between the first electrode plate **121** and the second electrode plate **122** within the battery during the charging and discharging. The electrolyte may be a non-aqueous organic electrolyte, which is a mixture of lithium salt and high-purity organic solvent. In addition, the electrolyte may be a polymer using a high-molecular electrolyte or a solid electrolyte, but the type of the electrolyte is not limited herein.

(26) As described above, an embodiment of the present invention may provide a secondary battery in which the safety vent and the current collector plate are electrically connected to each other through the electrode tab having a flat plate shape, and accordingly, the component resistance or

the internal resistance of the secondary battery may be minimized. Furthermore, an embodiment of the present invention may provide the secondary battery in which the output of the battery may be enhanced as the internal resistance is minimized, and the heat generation phenomenon of the battery may be reduced as well. Also, an embodiment of the present invention may provide the secondary battery in which since a current interrupting device, an insulating ring, or the like according to the related art is not required, the material costs of components are reduced.

(27) FIG. 2A is an enlarged cross-sectional view illustrating a cap assembly in a secondary battery according to an embodiment of the present invention, and FIG. 2B is a perspective view illustrating relationship between a current collector plate and an electrode tab.

(28) As illustrated in FIGS. 2A and 2B, the current collector plate **130** may be formed in an approximately circular flat plate shape, and may further include the recessed part **131** which has an approximately cross shape and is laser-welded to the first non-coating part **124** of the electrode assembly **120** after pressing the first non-coating part **124**. Also, the current collector plate **130** may include a central through-hole **132** formed at an approximately center and a plurality of peripheral through-holes **133** formed at a periphery thereof. As described above, the welding rod may enter through the central through-hole **132**, and the electrolyte may be injected into the electrode assembly **120** through the peripheral through-holes **133**.

(29) Also, the cap assembly **150** may include the safety vent **151** having an approximately circular flat plate shape, and the safety vent **151** may further include a vent groove **151a** formed in an approximately upper surface. The vent groove **151a** may have an approximately circular ring shape, and the thickness of the vent groove **151a** may be less than the thickness of a peripheral region of the vent groove **151a**. Also, the safety vent **151** may further include a bent part **151b** which is bent upward along an edge and coupled to the edge of the cap-up **152**. That is, the safety vent **151** and the cap-up **152** may be integrated with each other. To this end, the bent part **151b** of the safety vent **151** and the cap-up **152** may be welded to each other.

(30) In addition, a central region **151c** of the safety vent **151**, which is connected to the electrode tab **160**, may have a thickness greater than a thickness of a peripheral region thereof, and may have a shape protruding toward the electrode tab **160**. Thus, the central region **151c** of the safety vent **151** may be appropriately welded to the electrode tab **160**.

(31) Also, the cap-up **152** may include an edge flat part **152a**, a bent part **152b**, and a central flat part **152d**. The edge flat part **152a** may have a flat and approximately circular ring shape and may be coupled and electrically connected to the bent part **151b** of the safety vent **151**. Also, the bent part **151b** may be bent so as to be inclined upward from the edge flat part **152a** to the inside, and may include a peripheral through-hole **152c**. The peripheral through-hole **152c** may play a role in discharging an internal gas of the battery to the outside when the safety vent **151** ruptures, and may serve as a window enabling a laser beam for welding to arrive at the central region **151c** of the safety vent **151** in a case in which the central through-hole **152e** is not provided. The central flat part **152d** may extend horizontally from the bent part **152b** and spaced a predetermined distance from the safety vent **151**, and may include a central through-hole **152e** formed at an approximately center. The central through-hole **152e** may serve as a window enabling the laser beam to arrive at the central region **151c** of the safety vent **151** when the safety vent **151** and the electrode tab **160** are laser-welded to each other.

(32) Meanwhile, the electrode tab **160** may have an approximately flat top surface and an approximately flat bottom surface on the opposite side from the top surface. As described above, the electrode tab **160** may be interposed between the safety vent **151** and the current collector plate **130** while maintaining a flat shape. In embodiments, the thickness of the electrode tab **160** may be about 0.1 mm to about 0.5 mm, and the width may be about 10 mm to about 20 mm. Here, a central region of the top surface of the electrode tab **160** may be laser-welded to the central region of the safety vent **151**, and an edge region of the bottom surface of the electrode tab **160** may be welded to the current collector plate **130**. In addition, the electrode tab **160** is illustrated as an

approximately quadrangular plate shape in the drawing, but may have a circular disc shape.

(33) FIGS. 3A to 3F are views for explaining input/output of current, and various contact surfaces and welding surfaces in a secondary battery according to an embodiment of the present invention.

(34) FIG. 3A is a plan view of the cap-up **152**, and illustrates that a region X1, to or from which current is input or output, has an approximately circular ring shape. That is, the charging/discharging current may flow through the region X1 that has a donut shape indicated by hatching.

(35) FIG. 3B illustrates a contact surface X2 through the electrode tab **160** connected to the first current collector plate **130**. That is, the electrode tab **160** may be in contact with the central region **151c** of the safety vent **151** through a region X2 that has a circular shape indicated by hatching.

(36) FIG. 3C illustrates a welding surface X3 through the electrode tab **160** connected to the first current collector plate **130**. That is, the electrode tab **160** may be welded to the safety vent **151** through the region X3 that has a circular shape indicated by hatching. Here, the area of the welding surface X3 may be less than the area of the contact surface X2.

(37) FIG. 3D illustrates a contact surface X4 formed between the electrode tab **160** and the first current collector plate **130**. That is, the contact surface X4 between the electrode tab **160** and the first current collector plate **130** may be formed having an approximately quadrangular line shape. However, since a plurality of recessed parts **131** are formed in the first current collector plate **130**, the contact surface X4 is not formed in the recessed parts **131**.

(38) FIG. 3E illustrates a welding surface X5 formed between the electrode tab **160** and the first current collector plate **130**. That is, the welding surface X5 between the electrode tab **160** and the first current collector plate **130** may also be formed having an approximately quadrangular line shape. Here, the width of the welding surface X5 may be less than the width of the contact surface X4 described above.

(39) Finally, FIG. 3F illustrates an input or output region X6 of current, which is formed by welding the first current collector plate **130** and the first non-coating part **124**. That is, the input or output region X6 of the charging/discharging current is formed in a region corresponding to the recessed part **131** of the first current collector plate **130**, and may be formed having an approximately cross shape.

(40) With the above structure, the cap assembly **150** according to the present invention has component resistance or internal resistance lower than that of the related art. As one example, electric resistance of the cap-up **152** and the safety vent **151** is about 0.0826 mΩ, and electric resistance of the electrode tab **160** is 0.0636 mΩ. Also, resistance of the first current collector plate **130** is about 0.0149 mΩ. Thus, overall component resistance or internal resistance of the cap assembly **150** may be 0.01611 mΩ. The internal resistance of the cap assembly **150** according to an embodiment of the present invention is merely about 34% of the internal resistance of a cap assembly according to the related art. Thus, the embodiment of the present invention may provide the secondary battery in which the output is improved by minimizing the component resistance or internal resistance, and in which the material costs may be reduced. Also, as the internal resistance is minimized, the heat generation phenomenon of the secondary battery may be reduced as well.

(41) The above-described embodiment is merely one embodiment for embodying the secondary battery according to the present invention, and thus the present invention is not limited to the foregoing embodiment, and the technical spirits of the present invention include all ranges of technologies that may be variously modified by those of ordinary skill in the art, to which the present invention pertains, without departing from the subject matter of the present invention as set forth in the following claims.

Claims

1. A secondary battery comprising: a cylindrical can having an opening; an electrode assembly accommodated in the opening of the cylindrical can and wound in a cylindrical form; a current collector plate electrically connected to the electrode assembly, wherein the current collector plate further comprises a central through-hole formed in a central region; and a safety vent electrically connected to the current collector plate and sealing the opening of the cylindrical can, wherein an electrode tab, which has a flat top surface and a flat bottom surface on an opposite side from the flat top surface and has a thin plate shape whose length direction is perpendicular to a winding axis direction of the electrode assembly, is further interposed between the safety vent and the current collector plate, wherein the flat bottom surface extends over the central through-hole and at least a portion of the flat bottom surface connects to the current collector plate.
 2. The secondary battery of claim 1, wherein a central region of the flat top surface of the electrode tab is welded to the safety vent, and an edge region of the flat bottom surface of the electrode tab is welded to the current collector plate.
 3. The secondary battery of claim 1, wherein the current collector plate further comprises a peripheral through-hole formed on the outside of a central region.
 4. The secondary battery of claim 1, further comprising a cap-up, wherein the cap-up is electrically connected to an upper part of the safety vent.
 5. The secondary battery of claim 4, wherein the safety vent further comprises a bent part which is bent upward along an edge and coupled to an edge of the cap-up.
 6. The secondary battery of claim 4, wherein the cap-up comprises: an edge flat part electrically connected to the safety vent; a bent part bent upward from the edge flat part to the inside; and a central flat part extending horizontally from the bent part and spaced apart from the safety vent.
 7. The secondary battery of claim 6, wherein the bent part further comprises a peripheral through-hole.
 8. The secondary battery of claim 6, wherein the central flat part further comprises a central through-hole.
 9. The secondary battery of claim 1, wherein the safety vent includes a first portion and a second portion, the first portion being electrically connected to the current collector plate, and a first thickness of the first portion is greater than a second thickness of a second portion.
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